Week 10: Uncertainty

October 28, 2026

THE POINT OF THIS WEEK

An estimated coefficient is a best guess, not a fact. Fit the same model to a different sample of respondents and every $\hat{\beta}$ moves a little. A result reported without its uncertainty can't tell anyone whether "people prefer Fuji apples" is a real finding or noise from 12 pilot respondents.

So from now on every coefficient comes with a **standard error** and a **confidence interval**, and every plot of your results shows uncertainty bounds. The Pilot Analysis report requires both.

The second half of the week is a data skill you need to get there: reshaping your raw survey responses into the "long" `choiceData` format that `logitr` estimates models from.

WHERE UNCERTAINTY COMES FROM

The shape of the log-likelihood

Maximum likelihood finds the β that minimizes the negative log-likelihood, $-\ln(\mathcal{L}) = -\sum_j y_j \ln[P_j(\beta \mid x)]$. The *location* of the peak gives $\hat{\beta}$. The *curvature* of the peak gives the uncertainty. If the peak is flat, many other values of β fit the data almost as well.

Peak of $\ln(\mathcal{L})$	**$\hat{\beta}$ is**
Flat, broad	Less certain
Sharp, narrow	More certain

From the hessian to standard errors

The **hessian** is the matrix of second derivatives of $\ln(\mathcal{L})$ — it measures that curvature. Its negative inverse is the covariance matrix of $\hat{\beta}$:

$$\Sigma_\beta = -\left[\nabla^2_\beta \ln(\mathcal{L})\right]^{-1}$$

The diagonal of Σ_β holds the variances $\sigma_1^2, \sigma_2^2, \ldots$; their square roots are the **standard errors** reported next to each estimate. A 95% confidence interval is approximately $[\hat{\beta} - 2\sigma, \hat{\beta} + 2\sigma]$.

> Standard errors are the square roots of the diagonal of the negative inverse hessian. A sharper log-likelihood peak means bigger (more negative) hessian entries, smaller standard errors, and a tighter confidence interval.

```
hessian <- matrix(c(-6000, 60, 60, -700),
                  ncol = 2, byrow = TRUE)
covariance <- -1 * solve(hessian)
se <- sqrt(diag(covariance))

beta <- c(-0.4, 0.5)
beta - 2 * se   # lower bounds
beta + 2 * se   # upper bounds
```

SIMULATING UNCERTAINTY

Instead of the $\pm 2\sigma$ formula, take thousands of **draws** from a multivariate normal distribution centered on $\hat{\beta}$ with covariance Σ_β. Each row of draws is one plausible set of coefficients. The mean of the draws recovers $\hat{\beta}$, and the 2.5% and 97.5% quantiles give the 95% interval.

```
draws <- MASS::mvrnorm(10^4, beta, covariance)
mean(draws[, 1])
quantile(draws[, 1], c(0.025, 0.975))
logitr::ci(draws, level = 0.95)  # all columns
```

Why bother when the formula works? Because the draws carry through *any* calculation. Compute WTP or a market share from each row of draws and you get that quantity's uncertainty for free. That is how we'll do it in the coming weeks.

With an estimated `logitr` model

Function	**Returns**
`summary(model)`	Estimates, std. errors, z, p-values
`coef(model)`	The $\hat{\beta}$ vector
`vcov(model)`	The covariance matrix Σ_β
`MASS::mvrnorm(n, mu, Sigma)`	`n` draws of the coefficients
`ci(draws, level = 0.95)`	Mean, lower, upper for each column

```
coefs <- coef(model)
covariance <- vcov(model)
coef_draws <- as.data.frame(
  MASS::mvrnorm(10^4, coefs, covariance))
coef_ci <- ci(coef_draws, level = 0.95)
```

Plotting it

Plot the utility of each attribute level relative to its lowest level, with the interval around it (see `code/5.1-uncertainty.R` in `logitr-cars`):

Attribute type	**Geoms**
Continuous (price, mpg)	`geom_line()` + `geom_ribbon(alpha = 0.2)`
Categorical (powertrain)	`geom_point()` + `geom_errorbar()`

Map `ymin = lower` and `ymax = upper`, and give every panel the **same y-axis** so the attributes can be compared by eye.

TIDY DATA: "LONG" VS. "WIDE"

Tidy data is **"long"** format:

- Each **variable** has its own **column**
- Each **observation** has its own **row**

The quick test: **do the column names describe the values?** In a long table, `department`, `year`, and `rd_budget_mil` say what's in each column. In a wide table the names *are* values — `DOD`, `NASA`, `DOE` are levels of a department variable that got spread across columns.

Data	**Format**
`country, year, cases, population`	Long
`Film, Race, Female, Male`	Wide (`Female` and `Male` are values of gender)
`Film, Race, Gender, Word_Count`	Long

Reshaping with `tidyr`

```
# wide -> long
fed_spend_long <- fed_spend_wide %>%
  pivot_longer(
    names_to = "department",
    values_to = "rd_budget_mil",
    cols = DOD:Other)

# long -> wide
fed_spend_wide <- fed_spend_long %>%
  pivot_wider(
    names_from = department,
    values_from = rd_budget_mil)
```

Argument	**Says**
`cols`	Columns to stack
`names_to`	New column for the old names
`values_to`	New column for the values
`names_from`	Column that becomes names
`values_from`	Column that fills them

`cols` takes a range (`DOD:Other`) or everything except some columns (`-year`). The new names in `names_to` and `values_to` are quoted; the existing columns in `names_from` and `values_from` are not.

FROM PILOT RESPONSES TO `choiceData`

Your survey export has **one row per respondent**, with one column per choice question (`cbc_q1` … `cbc_q6`) holding the option they picked. `logitr` needs **one row per alternative, per question, per respondent**, with `choice` equal to 1 for the chosen alternative and 0 otherwise. Getting there is a `pivot_longer()`:

```
choice_data <- data %>%
  pivot_longer(
    cols = cbc_q1:cbc_q6,
    names_to = "qID",
    values_to = "choice") %>%
  mutate(qID = parse_number(qID),
         choice = parse_number(choice))
```

Then join the design (`choice_questions.csv`) by `respID` and `qID`, and recode with `choice = ifelse(choice == altID, 1, 0)`. The template walks through every step, including the filters that drop incomplete, speeding, and straight-lined responses.

Step	File in `template_conjoint_buttons`
Clean, build `choiceData`	`code/3_data_cleaning.R`
Estimate models	`code/4_modeling.R`

Open the downloaded `template_conjoint_buttons-main` folder in Positron, not the individual script, so the `file.path("data", ...)` paths resolve.

READING YOUR PILOT RESULTS

- **Sign first.** A negative price coefficient is expected. A dummy coefficient is utility *relative to the reference level* you left out.
- **Does the 95% interval include zero?** If it does, the data can't tell that attribute's effect apart from no effect.
- **Expect wide intervals.** A pilot has few respondents, so standard errors are large. Wide intervals are a reason to keep the attribute and collect more data, not proof it doesn't matter. Next week's power analysis is about choosing a sample size that shrinks them.
- $\pm 2\sigma$ is shorthand for $\pm 1.96\sigma$, so simulated quantiles and the formula agree to about two decimals.

ERRORS YOU WILL HIT THIS WEEK

Message	Usual cause
`exactly singular`	`solve()` on a hessian from an unidentified model, e.g. a dummy for every level
`NaNs produced`	Forgot the minus sign: `sqrt()` of negative variances
`not positive definite`	Passed the hessian to `mvrnorm()`, not the covariance
`not uniquely identified`	`pivot_wider()` found two values for one cell; an ID column is missing
`Can't combine`	`pivot_longer()` stacking text and number columns; narrow `cols`

BEFORE NEXT CLASS

- Pilot Analysis report — due **Nov 1**. Coefficient table *with standard errors*, and figures *with uncertainty bounds*.
- HW 9: Design of Experiment — due **Nov 2**.
- **Quiz 4** is at the start of class on Nov 4.

You should be able to:

- Explain why a flatter log-likelihood means a less certain estimate.
- Compute standard errors and a 95% CI from a hessian by hand in R.
- Simulate coefficient draws with `mvrnorm()` and summarize them with `ci()`.
- Plot estimated utilities with error bars or ribbons.
- Tell long from wide data, and convert between them with `pivot_longer()` and `pivot_wider()`.
- Turn your pilot survey responses into a long `choiceData` data frame and estimate a preliminary model with `logitr`.